



Standard XII » Mathematics » Combination

Question

A group consists of 4 girls & 7 boys, In how many ways can a team of 5 members be selected if the team has "at least one boy & one girl?"

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Solution



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At least one boy and one girl

Girls (out of 4)	Boys (out of 7)	Number of ways
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1	${}^4C_1 \times {}^7C_4$	$= 140$
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2	${}^4C_2 \times {}^7C_3$	$= 210$
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3	${}^4C_3 \times {}^7C_2$	$= 84$
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4	${}^4C_4 \times {}^7C_1$	$= 07$
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∴ Required number of ways

$$= 140 + 210 + 84 + 07 = 441$$

13. A and B can together finish a work 30 days. They worked together for 20 days and then B left. After another 20 days, A finished the remaining work. In how many days A alone can finish the work?
- Ⓐ 40
 - Ⓑ 50
 - Ⓒ 54
 - Ⓓ 60 

Answer: Option Ⓓ

Explanation:

$$(A + B)\text{'s 20 day's work} = \left(\frac{1}{30} \times 20 \right) = \frac{2}{3}.$$

$$\text{Remaining work} = \left(1 - \frac{2}{3} \right) = \frac{1}{3}.$$

Now, $\frac{1}{3}$ work is done by A in 20 days.

Therefore, the whole work will be done by A in $(20 \times 3) = 60$ days.

15. A train 110 metres long is running with a speed of 60 kmph. In what time will it pass a man who is running at 6 kmph in the direction opposite to that in which the train is going?

Ⓐ 5 sec

Ⓑ 6 sec ✓

Ⓒ 7 sec

Ⓓ 10 sec

Answer: Option Ⓑ

Explanation:

Speed of train relative to man = $(60 + 6)$ km/hr = 66 km/hr.

$$= \left(66 \times \frac{5}{18} \right) \text{m/sec}$$

$$= \left(\frac{55}{3} \right) \text{m/sec.}$$

$$\therefore \text{Time taken to pass the man} = \left(110 \times \frac{3}{55} \right) \text{sec}$$

9. A man can row at 5 kmph in still water. If the velocity of current is 1 kmph and it takes him 1 hour to row to a place and come back, how far is the place?

Ⓐ 2.4 km 

Ⓑ 2.5 km

Ⓒ 3 km

Ⓓ 3.6 km

Answer: Option Ⓐ

Explanation:

Speed downstream = $(5 + 1)$ kmph = 6 kmph.

Speed upstream = $(5 - 1)$ kmph = 4 kmph.

Let the required distance be x km.

$$\text{Then, } \frac{x}{6} + \frac{x}{4} = 1$$

$$\Rightarrow 2x + 3x = 12$$

$$\Rightarrow 5x = 12$$

$$\Rightarrow x = 2.4 \text{ km.}$$

If 4 men and 6 boys can do a work in 8 days and 6 men and 4 boys can do the same work in 7 days, then how many days will 5 men and 4 boys take to do the same work?

Given:

4 men and 6 boys can do a work in 8 days

6 men and 4 boys can do the same work in 7 days

Concept used:

Time = Work/Efficiency

Calculation:

Let, M = Men; B = Boys

$$(4M + 6B) \times 8 = (6M + 4B) \times 7$$

$$\Rightarrow 32M + 48B = 42M + 28B$$

$$\Rightarrow 42M - 32M = 48B - 28B$$

$$\Rightarrow 10M = 20B$$

$$\Rightarrow M/B = 2/1$$

So, the efficiency of one man = 2

Efficiency of one boy = 1



$$\text{Total work} = [(4 \times 2) + (6 \times 1)] \times 8 = 14 \times 8 = 112$$

$$\text{Efficiency of 5 men and 4 boys} = (5 \times 2) + (4 \times 1) = 10 + 4 = 14$$

Required time to finish the work = $112/14 = 8$ days

$\therefore$ The 5 men and 4 boys will take 8 days to do the same work

Question

James' father was 30 years old when he was born. His mother's age was 24 when his sister who is 5 years younger to him, was born. What is the difference between the age of James' father and mother?

Solution

The correct option is **D** 11

Jame's age = $F - 30$

Sister's age = $F - 35$

According to Question

$M = 24 + \text{Sister's age}$

$M = 24 + F - 35$

$F - M = 11$

There are 20 students with an average height of 105 cms in a class. Then 10 students with an average height of 120 cms join the class. What will be the average height of the class now?

20 students average height is 105cms

sum of heights/ 20 =105cms

sum of heights of 20 students= $105 \times 20 = 2100$

10 students of average height 120 added

so their, sum of 10 students height / 10 = 120

sum of heights of 10 students = 1200cms

By adding heights of 20 and 10 students , $1200 + 2100 = 3300$

average of 30 students = $3300 / 30 = 110$ cms

Answer is 110 cms

Question

In a test containing of 80 questions carrying one mark each, Arpita answered 65 % of the first 40 questions correctly. What percent of the other 40 questions does she need to answer correctly for her grade in the entire exam to be 75 %?

A 60

B 80

C 85

D 75

Correct Answer

Solution



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Let the required per cent of question be x

Then 65% of $40 + x\%$ of $40 = 75\%$ of 80

$$\Rightarrow \frac{65}{100} \times 40 + \frac{x}{100} \times 40 = \frac{75}{100} \times 80$$

$$x = \left(\frac{75 \times 80}{100} - \frac{65 \times 40}{100} \right) \times \frac{100}{40} \%$$

$$= (60 - 26) \times \frac{100}{40} \% = 85\%$$

Q: A thief is noticed by a policeman from a distance of 200 m. The thief starts running and the policeman chases him. The thief and the policeman run at the rate of 10 km and 11 km per hour respectively. What is the distance between them after 6 minutes?

244 

- A) 100 m
- B) 150 m
- C) 190 m
- D) 200 m

Answer: A) 100 m

Explanation:

Relative speed of the thief and policeman
 $= (11 - 10) \text{ km/hr} = 1 \text{ km/hr}$

Distance covered in 6 minutes $= (1/60) \times 6 \text{ km} = 1/10 \text{ km} = 100 \text{ m}$

Therefore, Distance between the thief and policeman $= (200 - 100) \text{ m} = 100 \text{ m}.$

Raju walks 80 m towards south. Then, turns to his right and starts walking straight till he completes another 80 m. Then, again turning to his left, he walks for 60 m. He then turns to his left and walks for 80 m. How far is he from his initial position?

☐ A 240 m

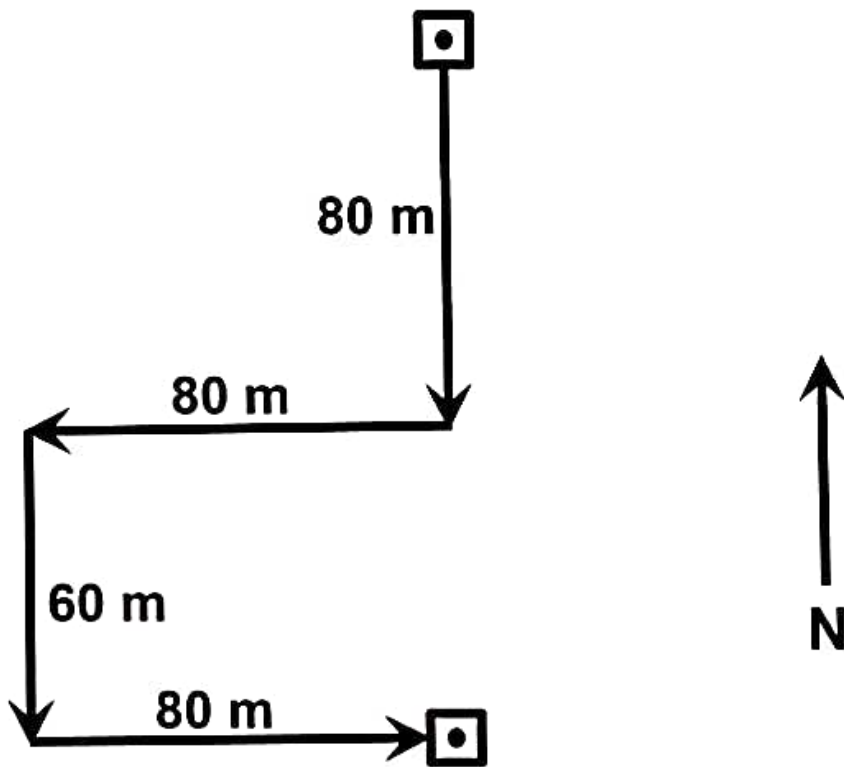
☐ B 120 m

☒ C 140 m



Solution

The correct option is **C** 140 m



$\therefore$ Required distance = $80 + 60 = 140$ m

Q9 Find out the wrong number in the series.

7, 8, 18, 57, 228, 1165, 6996

A 8

B 18

C 57

D 228

Answer: 228

Explanation

*Let the given numbers be A ,
 B , C , D , E , F , G .*

Then, A ,

$$A \times 1 + 1,$$

$$B \times 2 + 2,$$

$$C \times 3 + 3,$$

$$D \times 4 + 4,$$

$$E \times 5 + 5,$$

*$F \times 6 + 6$ are the required
numbers.*

Clearly, 228 is wrong.

Q10 Find the odd man out.

253, 136, 352, 324, 631, 244

☒ A 324

☐ B 136

☐ C 352

☐ D 631

Answer: 324

Explanation

*The Sum of the digits in each number,
Except 324 is 10.*